

SANTIAGO DE CUBA, CubaWMO: **782640**Lat: **19.970N**Lon: **75.835W**Elev: **249**StdP: **14.56**Time Zone: **-5.00 (CUB)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	65.9	67.6	53.6	61.7	73.2	55.9	67.1	73.2	21.2	75.1	19.3	75.8	9.5	0	0.342

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.7	89.8	76.8	89.2	77.0	88.1	77.1	80.9	85.5	80.1	85.3	79.4	85.0	10.0	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.1	157.7	84.0	78.9	151.2	83.6	77.7	145.1	83.2	44.9	85.3	43.9	85.5	43.2	85.3	85.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.1	15.7	13.9	63.0	94.7	2.2	3.1	61.4	96.9	60.1	98.7	58.8	100.5	57.2	102.7		
(5)			57.5	81.8	2.4	1.8	55.8	83.2	54.4	84.2	53.1	85.3	51.3	86.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	79.2	76.0	76.3	76.8	78.1	79.4	80.8	81.8	82.3	81.7	80.6	79.1	77.5
	DBStd	2.75	2.18	1.99	1.92	1.51	1.61	1.44	1.22	1.35	1.41	1.60	1.97	2.16
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	10660	806	737	830	842	911	923	986	1000	950	949	873	853
	CDD65	5185	341	317	365	392	446	473	521	535	500	484	423	388
	CDH74	47192	2223	2259	2726	3235	4122	4946	5752	5987	5299	4537	3347	2759
	CDH80	14660	435	434	537	744	1170	1667	2169	2314	1968	1503	987	732
Wind	WSAvg	7.0	8.0	7.8	8.0	7.3	6.5	6.0	6.3	6.6	6.2	6.4	7.5	7.6
Precipitation	PrecAvg	44.3	1.6	1.6	2.2	3.0	7.0	3.7	3.2	3.6	4.8	7.4	3.8	1.6
	PrecMax	100.7	9.8	7.7	5.9	10.5	19.0	10.2	10.0	9.0	19.0	53.5	12.6	5.2
	PrecMin	25.1	0.0	0.0	0.1	0.0	0.2	0.4	0.2	1.5	0.9	1.5	0.3	0.0
	PrecStd	15.0	2.2	1.7	1.6	2.1	5.1	2.3	1.9	1.7	3.2	8.8	2.8	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	86.3	86.3	86.5	86.4	88.1	89.3	90.0	91.5	91.1	89.8	89.2	88.0
		MCWB	71.3	71.1	71.4	74.0	76.3	77.3	76.9	76.8	76.5	76.5	74.4	73.4
	2%	DB	84.5	84.5	84.6	85.7	86.4	88.0	89.4	89.8	89.5	88.2	87.6	86.2
		MCWB	71.9	72.0	72.6	74.1	76.1	77.4	77.2	77.3	77.4	76.9	75.0	73.5
	5%	DB	83.8	83.9	84.1	84.4	85.9	87.5	88.1	88.9	88.1	87.6	86.2	84.9
		MCWB	72.0	72.1	72.7	73.9	76.0	77.3	77.3	77.4	77.4	77.0	75.2	73.3
	10%	DB	82.5	82.6	82.7	83.9	84.6	86.3	87.7	87.9	87.7	86.3	85.1	84.1
		MCWB	72.0	72.0	72.1	73.7	75.6	77.2	77.4	77.6	77.4	76.8	75.0	73.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	78.0	78.9	79.3	79.7	80.9	81.8	81.4	81.2	81.6	81.1	80.5	79.2
		MCDB	82.1	81.8	82.5	83.1	84.2	85.0	86.7	86.8	85.7	85.0	84.0	83.9
	2%	WB	75.9	76.4	77.1	77.7	79.2	80.4	80.0	80.1	80.1	79.7	78.9	77.4
		MCDB	81.1	81.0	81.6	82.6	83.5	84.5	86.2	86.5	86.0	84.8	83.6	82.7
	5%	WB	74.5	74.6	75.3	76.3	78.3	79.3	79.2	79.2	79.1	78.8	77.8	75.9
		MCDB	80.7	80.4	81.2	82.1	83.1	84.4	85.7	86.0	85.5	84.4	83.2	81.8
	10%	WB	73.3	73.5	74.1	75.3	77.3	78.6	78.6	78.7	78.5	78.0	76.7	74.7
		MCDB	80.3	80.3	80.6	81.7	82.8	84.2	85.4	85.7	85.1	84.0	82.6	81.0
Mean Daily Temperature Range	5% DB	MDBR	13.9	14.2	13.5	12.3	11.7	11.7	12.5	12.7	12.7	12.4	13.2	14.0
		MCDBR	15.2	14.9	14.1	12.9	12.5	12.8	13.2	13.6	13.5	13.7	14.2	15.0
		MCWBR	7.5	7.5	7.0	6.7	6.5	6.8	7.0	6.7	6.4	6.2	6.5	7.1
	5% WB	MCDBR	13.4	13.1	11.8	11.5	10.9	11.3	12.2	12.2	12.3	11.7	12.0	13.3
	MCWBR	8.3	8.1	7.3	7.3	7.0	7.1	7.3	6.9	6.8	6.7	7.1	7.7	
Clear-Sky Solar Irradiance	taub	0.374	0.380	0.386	0.422	0.449	0.484	0.492	0.462	0.424	0.401	0.388	0.382	
	taud	2.520	2.494	2.479	2.388	2.343	2.232	2.194	2.312	2.437	2.508	2.505	2.513	
	Ebn at Noon	286	291	292	281	270	259	257	266	277	281	280	280	
	Edh at Noon	31	34	36	39	41	45	47	42	36	33	31	31	
All-Sky Solar Radiation	RadAvg	1427	1696	1898	2004	1915	1923	2014	1951	1808	1549	1427	1350	
	RadStd	93	92	93	104	113	116	99	64	116	130	101	83	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	+2.99	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page